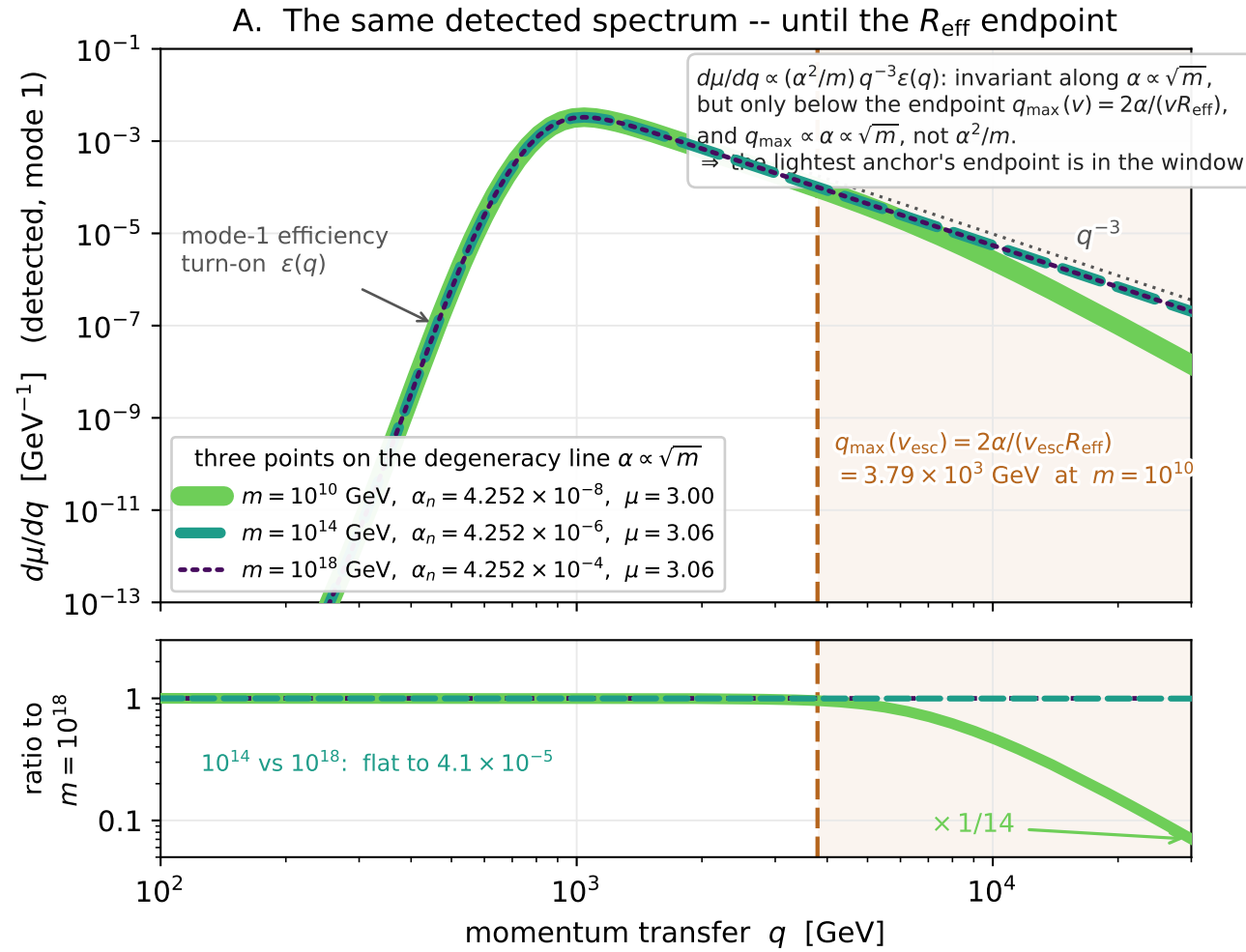
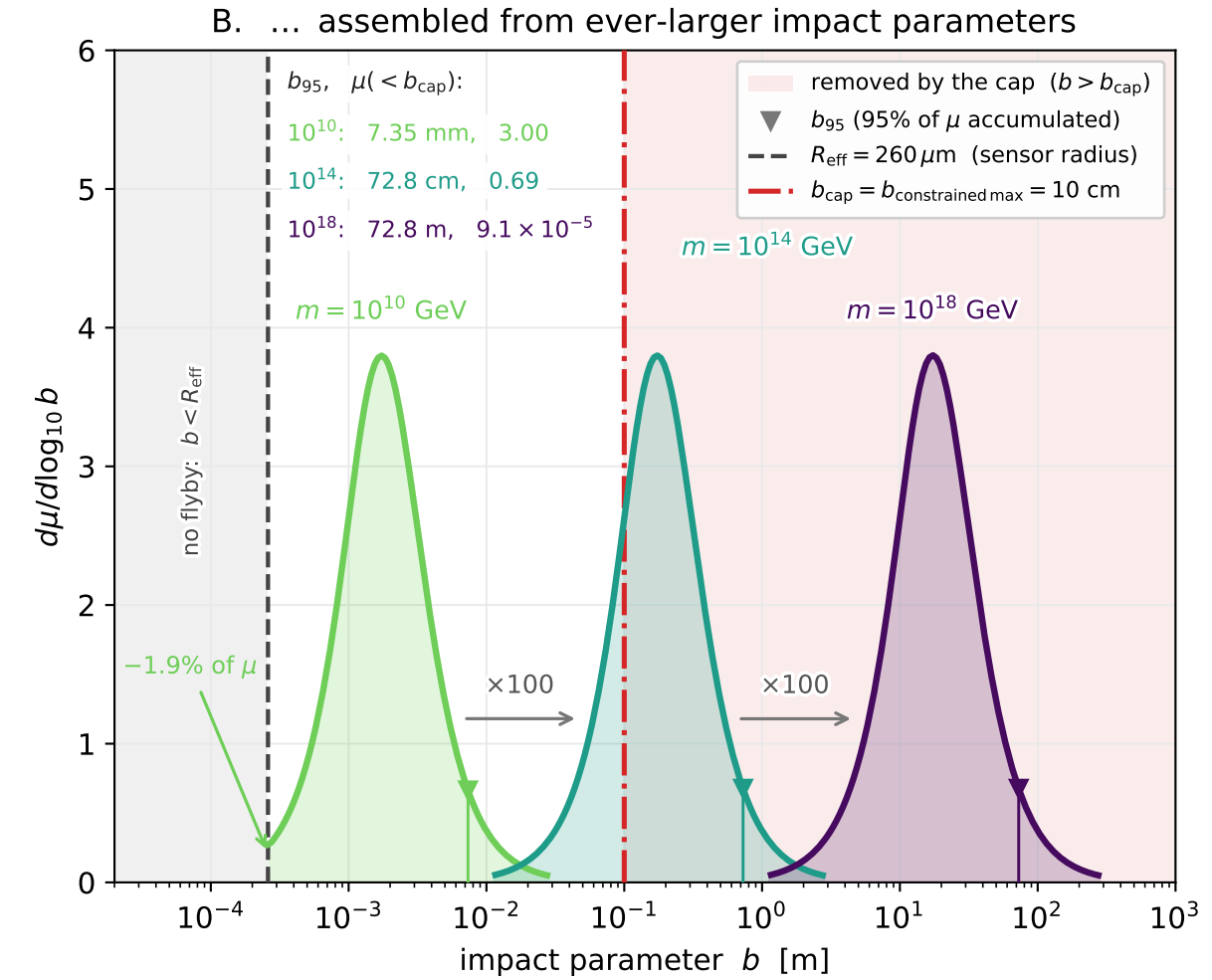


The α^2/m degeneracy of the long-range ($1/r^2$) interaction: broken at low mass by the sensor radius, closed at high mass by the impact-parameter cap



$q_{\text{th}} = 100$ GeV $T = 0.791 \times 10^6$ s $f_X = 0.1$ $R_{\text{eff}} = 260 \mu\text{m}$



anchor $\alpha_n = 4.252 \times 10^{-8}$ at $m = 10^{10}$ GeV, $\times 100$ per $\times 10^4$ in m

DOMAIN OF VALIDITY along $\alpha \propto \sqrt{m}$: the spectra coincide to $< 10^{-3}$ across 10^2 - 3×10^4 GeV for $m \gtrsim 6.5 \times 10^{12}$ GeV (μ to $< 10^{-3}$ for $m \gtrsim 2.0 \times 10^{11}$ GeV); below that the R_{eff} endpoint breaks the degeneracy -- by 1.9% in μ at $m = 10^{10}$ GeV, which does not close the island; the b_{cap} of panel B does.